CHM111: General Chemistry I: Practice Exam
Dr. Deskins
Name:______________________________

This practice exam covers Chapters 10, 11 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Calculate the heat required to completely vaporize 37.6-g of methanol (CH_3OH) at its normal boiling point of 65°C. $\Delta H_{vap} = 35.2\,\text{kJ/mol}$, $M = 32.042\,\text{g/mol}$.

   ANSWER: ______________________________

2. Why does an intermolecular potential-energy curve rise very steeply at separations shorter than the equilibrium distance? Justify.

   ANSWER: ______________________________

3. When a narrow copper capillary tube is dipped into mercury, does the mercury rise or become depressed in the tube? Justify using cohesive and adhesive forces.

   ANSWER: ______________________________

4. How many grams of glucose ($C_6H_{12}O_6$) must be dissolved in 500. g of water to raise the boiling point by 1.02 °C? $K_b(H_2O) = 0.512\,°\text{C kg/mol}$.

   ANSWER: ______________________________

5. How many hydrogen-bond donors does one molecule of HF have?

   ANSWER: ______________________________

6. What is the coordination number of each atom in a body-centered cubic (BCC) structure?

   ANSWER: ______________________________

7. Rank the following noble gases from highest to lowest boiling point: Kr, Xe, Ar, He. All noble gases interact only via London dispersion forces.

   ANSWER: ______________________________

8. An X-ray diffraction experiment on a crystal produces a strong reflected beam under the conditions below. A crystal with plane spacing 295 pm produces the first-order diffraction maximum at a glancing angle of $\theta = 22.8°$. Determine the wavelength of the incident X-rays. Determine the requested quantity. A correct answer requires a justification based on the underlying physical cause rather than a memorized relationship.

ANSWER: ______________________________

9. The normal boiling point of benzene (C_6H_6) is 80.10°C ($P_{\text{vap}} = 1.000\,\text{atm}$) and its molar enthalpy of vaporization is 30.8 kJ/mol. Calculate the vapor pressure of benzene at 53.10°C.

ANSWER: ______________________________

10. Calculate the heat required to completely vaporize 16.8-g of ethanol (C_2H_5OH) at its normal boiling point of 78°C. $\Delta H_{\text{vap}} = 38.6\,\text{kJ/mol}$, $M = 46.068\,\text{g/mol}$.

ANSWER: ______________________________

11. For one mole of ammonia (NH_3), which requires the most energy: melting, boiling, or subliming? Justify.

ANSWER: ______________________________

12. Calculate the osmotic pressure of a $0.500\,\text{M}$ aqueous solution of urea (CH_4N_2O) at 25°C.

ANSWER: ______________________________

13. Ti crystallizes in a hexagonal close-packed structure with unit cell edge length $295.1\,\text{pm}$. Calculate the density of Ti.

ANSWER: ______________________________

14. Rank the following liquids from highest to lowest viscosity at 20°C: honey, CH_3COCH_3, H_2O, motor oil.

ANSWER: ______________________________

15. How many grams of C_3H_6O can vaporize when 22.7-kJ of heat is absorbed during vaporization at 56°C? Assume $\Delta H = 518\,\mathrm{J/g}$.

ANSWER: ______________________________

16. An unknown solid has a melting point of -57°C, is a non-conductor as a solid, and is a non-conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: ______________________________

17. Dissolving lithium chloride (LiCl) is exothermic ($\Delta H_{\mathrm{sol}} = -37.0\,\mathrm{kJ/mol}$). Identify the two factors that decide whether it dissolves spontaneously. Explain.

ANSWER: ______________________________

18. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.150 M KCl, 0.100 M $C_6H_{12}O_6$, 0.150 M Na_2SO_4. Justify each classification.

ANSWER: ______________________________

19. Equal masses (1.00 g each) of Na_2SO_4, CH_3OH, $CaCl_2$ are each dissolved in 1.00 kg of water. Which solution has the greatest boiling point elevation?

ANSWER: ______________________________

20. Fe crystallizes in a body-centered cubic structure with unit cell edge length 286.6 pm. Calculate the density of Fe.

ANSWER: ______________________________

21. Which substance has the lower vapour pressure at the same temperature: 1-propanol ($CH_3CH_2CH_2OH$) or propanal (CH_3CH_2CHO)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: ______________________________

22. Rank the following substances from highest to lowest vapor pressure at room temperature: H_2O, C_2H_5OH, $(C_2H_5)_2O$. For credit you must justify your answer.

ANSWER: ______________________________

23. The following information is available for silver. Determine the density of silver and justify your answer.
unit-cell edge length = 408.53 pm
molar mass = 107.868 $g\,mol^{-1}$
atomic radius = 144 pm
crystal structure = FCC
atoms per unit cell = 4
packing efficiency = 74%

ANSWER: ____________________________

24. Calculate the osmotic pressure of a 0.500 M aqueous solution of glucose ($C_6H_{12}O_6$) at 37°C.

ANSWER: ____________________________

25. Classify a solid molecular sugar (sucrose) (sucrose ($C_{12}H_{22}O_{11}$)) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: ____________________________

26. The molar enthalpy of sublimation of benzene (C_6H_6) is 40.67 kJ/mol and its molar enthalpy of fusion is 9.87 kJ/mol. Calculate the molar enthalpy of vaporization.

ANSWER: ____________________________

27. Calculate the osmotic pressure of 0.90% saline at 25°C (modeled as a 0.154 M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: ____________________________

28. Classify NaCl (NaCl) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ____________________________

29. Calculate the molar solubility of CH_4 (CH_4) in water at 25°C when the partial pressure of CH_4 above the solution is 1.0 atm. Use $K_H = 1.4 \times 10^{-3}$ mol/(L·atm) (Henry's law: $C = K_H P$).

ANSWER: ____________________________

30. Calculate the vapor pressure of a solution prepared by dissolving 54.0 g of urea (CH_4N_2O) in 100. g of water at 60°C. The vapor pressure of pure water at 60°C is 149.4 mmHg.

ANSWER: ______________________________

31. A plant root cell (internal particle concentration $\approx$ 0.300 osmol/L) is placed in a 0.500 M solution of sucrose ($C_{12}H_{22}O_{11}$). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: ______________________________

32. Iron crystallizes in a BCC unit cell with an atomic radius of 126 pm. Calculate the edge length of the unit cell in pm.

ANSWER: ______________________________

33. A solution of N_2 (N_2) in water at 25°C is prepared at a partial pressure of 1.0 atm. If the partial pressure is increased to 5.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: ______________________________

34. Which substance has stronger attractions between neighbouring molecules: water (H_2O) or decane ($C_{10}H_{22}$)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: ______________________________

35. You must prepare a solution for a precise freezing-point-depression measurement. Which concentration unit should express the concentration, and why?

ANSWER: ______________________________

36. How does increasing pressure affect the solubility of NaCl in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

37. The normal boiling point of benzene (C_6H_6) is 80.10°C ($P_{\text{vap}} = 1.000\,\text{atm}$) and its molar enthalpy of vaporization is 30.8 kJ/mol. Calculate the vapor pressure of benzene at 60.10°C.

ANSWER: ______________________________

38. Two pure substances, HF and HCl, are compared. Predict which requires the higher temperature to boil. Justify your prediction. Unsupported trend statements receive no credit.

ANSWER: ______________________________

39. List every type of intermolecular force present in pure $CHCl_3$. Justify your answer.

ANSWER: ______________________________

40. A solution is prepared by dissolving 1.00 g of lysozyme in enough water to make 200. mL of solution. The osmotic pressure at 25°C is 8.60×10^{-3} atm. Calculate the molar mass of lysozyme. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathscr{M} = (m/V)/M$.

ANSWER: ______________________________

41. Calculate the boiling point of a solution prepared by dissolving 25.0 g of NaCl (NaCl) in 100. g of water. $K_b(H_2O) = 0.512°C\,\text{kg/mol}$. Assume complete dissociation ($i = 2$).

ANSWER: ______________________________

42. Calculate the osmotic pressure of brackish water at 25°C (modeled as a 0.0500 M electrolyte solution with $i = 1$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: ______________________________

43. Graphite and graphene both consist entirely of carbon. Determine which material would be expected to conduct electricity more effectively within its carbon layer(s). State your answer and justify it.

ANSWER: ______________________________

44. A solution is prepared by dissolving 25.0 g of glucose ($C_6H_{12}O_6$, $M = 180.16$ g/mol) in 500. g of water. Calculate the molality of the solution.

ANSWER: ______________________________

45. What is the coordination number of each atom in a hexagonal close-packed (HCP) structure?

ANSWER: ______________________________

46. Classify a solid composed of potassium and iodine (KI) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: ______________________________

47. Classify paint as a type of colloid: identify its dispersed phase and dispersion medium, then name the colloid type.

ANSWER: ______________________________

48. Diamond and solid CO_2 are both solids containing carbon. Determine which requires the higher temperature to melt or decompose into a liquid phase under sufficient pressure. State your answer and justify it.

ANSWER: ______________________________

49. Calculate the mole fractions of ethanol (C_2H_5OH) and water in a solution prepared by mixing 6.40 g of ethanol with 72.1 g of water.

ANSWER: ______________________________

50. The following information is available for gold. Determine the density of gold and justify your answer.
melting point = 1064.2 °C
coordination number = 12
atoms per unit cell = 4
unit-cell edge length = 407.82 pm
crystal structure = FCC
molar mass = 196.967 g mol^{-1}

ANSWER: ______________________________

51. Calculate the mole fractions of acetone (C_3H_6O) and water in a solution prepared by mixing 18.4 g of acetone with 100. g of water.

ANSWER: ______________________________

52. What is the packing efficiency (percent of space occupied by atoms) of a face-centered cubic (FCC) unit cell?

ANSWER: ______________________

53. How many grams of CO_2 can sublime when 10.0-kJ of heat is absorbed during sublimation at -78.5°C? Assume $\Delta H = 571\,\mathrm{J/g}$.

ANSWER: ______________________

54. A mixture contains visible particles that settle on standing. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________

55. A solution is prepared by dissolving 0.500 g of albumin in enough water to make 100. mL of solution. The osmotic pressure at 25°C is 1.90×10^{-3} atm. Calculate the molar mass of albumin. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathscr{M} = (m/V)/M$.

ANSWER: ______________________

56. Calculate the heat required to warm 24.2-g of ice from $-21°\mathrm{C}$ to $0°\mathrm{C}$. $c_{\mathrm{ice}} = 2.09\,\mathrm{J\ g^{-1}°C^{-1}}$.

ANSWER: ______________________

57. Calculate the heat required to completely vaporize 14.6-g of methanol (CH_3OH) at its normal boiling point of 65°C. $\Delta H_{\mathrm{vap}} = 35.2\,\mathrm{kJ/mol}$, $M = 32.042\,\mathrm{g/mol}$.

ANSWER: ______________________

58. Rank these primary amines from most to least soluble in water: $C_2H_5NH_2$, $C_3H_7NH_2$, CH_3NH_2, $C_6H_{13}NH_2$. Explain the trend.

ANSWER: ______________________

59. In an ionic crystal a cation (Mg^{2+}, radius 72 pm) sits among anions (O^{2-}, radius 140 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: ______________________________

60. What is the coordination number of each atom in a face-centered cubic (FCC) structure?

ANSWER: ______________________________

61. Calculate the heat absorbed when 14.0-g of benzene (C_6H_6) melts completely at its melting point of 6°C. $\Delta H_{\text{fus}} = 9.87\,\text{kJ/mol}$.

ANSWER: ______________________________

62. Equal masses (1.00 g each) of KNO_3, $C_3H_8O_3$, NaCl are each dissolved in 1.00 kg of water. Which solution has the greatest freezing point depression?

ANSWER: ______________________________

63. Rank the following substances in order of lowest to highest boiling point: CH_4, CH_3SH, CH_3Cl, CH_3OH. Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: ______________________________

64. A sample of iron has the properties listed below. Determine its density and justify your answer.
electron configuration = $[\text{Ar}]3d^6 4s^2$
molar mass = $55.845\,\text{g}\,\text{mol}^{-1}$
packing efficiency = 68%
atoms per unit cell = 2
unit-cell edge length = 286.65 pm

ANSWER: ______________________________

65. Rank the following equimolal aqueous solutions from smallest to largest osmotic pressure: CH_4N_2O, Na_2SO_4, $MgBr_2$. All solutions have the same molality.

ANSWER: ______________________________

66. Calculate the heat absorbed when 57.7-g of iron (Fe) melts completely at its melting point of 1538°C. $\Delta H_{\text{fus}} = 13.81\,\text{kJ/mol}$.

ANSWER: ______________________________

67. A solution is prepared by dissolving 25.0 g of urea (CH_4N_2O, $M = 60.06\,\text{g/mol}$) in 200. g of water. Calculate the mole fraction of urea.

ANSWER: ______________________________

68. Calculate the molar solubility of CO_2 (CO_2) in water at 25°C when the partial pressure of CO_2 above the solution is 0.50 atm. Use $K_H = 3.4 \times 10^{-2}\,\text{mol/(L·atm)}$ (Henry's law: $C = K_H P$).

ANSWER: ______________________________

69. Why does the temperature stay constant along the melting plateau of a heating curve even though heat is continuously added? Justify.

ANSWER: ______________________________

70. The lattice energy of KCl (KCl) is $+717\,\text{kJ/mol}$ (energy to separate ions) and the hydration energy is $-681\,\text{kJ/mol}$. Calculate ΔH_{sol}. Is dissolution endothermic or exothermic?

ANSWER: ______________________________

71. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.150 M $MgCl_2$, 0.100 M $CaCl_2$, 0.200 M $C_{12}H_{22}O_{11}$. Justify each classification.

ANSWER: ______________________________

72. The vapor pressure of chloroform ($CHCl_3$) is 0.2613 atm at 20.00°C and 1.000 atm at 61.00°C. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________________

73. The normal boiling point of ethanol (C_2H_5OH) is 78.40°C ($P_{\text{vap}} = 1.000\,\text{atm}$) and its molar enthalpy of vaporization is 38.6 kJ/mol. Calculate the vapor pressure of ethanol at 47.40°C.

ANSWER: ______________________

74. One of H_2S and H_2O has a substantially higher boiling point than the other. Determine which one and justify your answer.

ANSWER: ______________________

75. An ionic solid has a face-centered cubic array of anions with cations in all of the octahedral holes. Name this structure type (with a common example) and justify.

ANSWER: ______________________

76. A beam of light is passed through muddy water. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________

77. How many grams of H_2O can melt when 14.1-kJ of heat is absorbed during melting at 0°C? Assume $\Delta H = 334\,\text{J/g}$.

ANSWER: ______________________

78. Both CCP and HCP are close-packed. State the coordination number and packing efficiency they share, and the single feature that distinguishes them. Justify.

ANSWER: ______________________

79. Classify each as crystalline or amorphous: a quartz crystal and rubber. Justify using their internal order.

ANSWER: ______________________

80. Solid ammonia (NH_3) has its triple point at -77.7°C and 0.06 atm. It is held at a constant external pressure of 0.354 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: ______________________

81. Acetic acid (CH_3COOH, density 1.05 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: ______________________________

82. Identify the dominant type of intermolecular force present in pure Br_2. You must justify your answer.

ANSWER: ______________________________

83. Calculate the heat absorbed when 58.6-g of ethanol (C_2H_5OH) melts completely at its melting point of -114°C. $\Delta H_{\text{fus}} = 4.6\,\text{kJ/mol}$.

ANSWER: ______________________________

84. An unknown solid has the following properties: variable melting point, malleable/ductile, conducts electricity as solid. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: ______________________________

85. Describe the layer stacking sequence of hexagonal closest packing (HCP) and explain what makes it distinct. Justify.

ANSWER: ______________________________

86. For iodine (I_2): normal melting point 114°C, normal boiling point 184°C, triple point (114°C, 0.117 atm), critical point (546°C, 116 atm). What phase is present at 135°C and 0.0167 atm? Justify your answer.

ANSWER: ______________________________

87. For ammonia (NH_3), $\Delta H_{\text{fus}} = 5.66$ kJ/mol and $\Delta H_{\text{vap}} = 23.4$ kJ/mol. Explain why boiling absorbs so much more energy than melting.

ANSWER: ______________________________

88. Explain how a single crystal illuminated by monochromatic X-rays can produce many distinct diffraction peaks. Justify.

ANSWER: ______________________________

89. Which substance has the higher normal boiling point: acetone (CH_3COCH_3) or propan-2-ol ($(CH_3)_2CHOH$)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: ______________________________

90. When a narrow glass capillary tube is dipped into ethanol, does the ethanol rise or become depressed in the tube? Justify using cohesive and adhesive forces.

ANSWER: ______________________________

91. Calculate the vapor pressure of a solution prepared by dissolving 36.0 g of glucose ($C_6H_{12}O_6$) in 200. g of water at 80°C. The vapor pressure of pure water at 80°C is 355.1 mmHg.

ANSWER: ______________________________

92. A solution of benzene and toluene is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

93. What is the coordination number of each atom in a simple cubic (SC) structure?

ANSWER: ______________________________

94. Two isomers share the molecular formula C_8H_{18}: n-octane (straight-chain (elongated)) and 2,2,4-trimethylpentane (highly branched (compact)). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: ______________________________

95. Use the Born-Haber cycle to calculate ΔH_f° of NaF. $\Delta H_{\text{sub}}(\text{Na}) = 108\,\text{kJ/mol}$, $IE_1(\text{Na}) = 496\,\text{kJ/mol}$, $\frac{1}{2}D(\text{F}_2) = 79\,\text{kJ/mol}$, $EA(\text{F}) = -328\,\text{kJ/mol}$, $U(NaF) = -929\,\text{kJ/mol}$.

ANSWER: ______________________________

96. Why does the temperature stay constant along the melting plateau of a heating curve even though heat is continuously added? Justify.

ANSWER: ______________________________

97. Use the Born-Haber cycle to calculate ΔH_f° of NaF. $\Delta H_{\text{sub}}(\text{Na}) = 108\,\text{kJ/mol}$, $IE_1(\text{Na}) = 496\,\text{kJ/mol}$, $\frac{1}{2}D(\text{F}_2) = 79\,\text{kJ/mol}$, $EA(\text{F}) = -328\,\text{kJ/mol}$, $U(NaF) = -929\,\text{kJ/mol}$.

ANSWER: ______________________________

98. Nickel crystallizes in a FCC unit cell with an atomic radius of 124 pm. Calculate the edge length of the unit cell in pm.

ANSWER: ______________________________

99. Rank the following equimolal aqueous solutions from smallest to largest osmotic pressure: CH_4N_2O, $MgBr_2$, Na_2SO_4. All solutions have the same molality.

ANSWER: ______________________________

100. A 0.100 m aqueous solution of $MgCl_2$ ($MgCl_2$) shows a freezing point depression of $0.521\,^\circ\text{C}$ ($K_f = 1.86^\circ\text{C kg/mol}$). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: ______________________________

101. Calculate the osmotic pressure of a 0.100 M aqueous solution of urea (CH_4N_2O) at 25°C.

ANSWER: ______________________________

102. When heated, a quartz crystal melts sharply at one temperature, whereas window glass gradually softens over a range. Which is crystalline and which is amorphous? Justify.

ANSWER: ______________________________

103. A 25.6-g sample of liquid water at 100°C absorbs 31.6 kJ of heat. Determine the final state and temperature. ($\Delta H_{\text{vap}} = 2260$ J/g.)

ANSWER: ______________________________

104. The vapor pressure of diethyl ether ($(C_2H_5)_2O$) is 0.173 atm at $-10.00°C$ and 1.000 atm at $34.50°C$. Calculate ΔH_{vap} in kJ/mol. $R = 8.314\,J\,mol^{-1}K^{-1}$.

ANSWER: ______________________________

105. The normal boiling point of acetone (CH_3COCH_3) is $56.00°C$ ($P_{vap} = 1.000\,atm$) and its molar enthalpy of vaporization is 31.3 kJ/mol. Calculate the vapor pressure of acetone at $24.00°C$.

ANSWER: ______________________________

106. Two isomers share the molecular formula C_6H_{14}: 2,3-dimethylbutane (compact/branched) and n-hexane (straight-chain (elongated)). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: ______________________________

107. Explain how a single crystal illuminated by monochromatic X-rays can produce many distinct diffraction peaks. Justify.

ANSWER: ______________________________

108. Classify glucose ($C_6H_{12}O_6$) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

109. Two substances have intermolecular potential-energy curves of identical shape but different well depths. Which forms the solid with the higher melting point? Justify.

ANSWER: ______________________________

110. On the P–T phase diagram of sodium chloride (NaCl), the solid–liquid boundary has a positive slope. Which is denser, the solid or the liquid? Justify.

ANSWER: ______________________________

111. Iron crystallizes in a BCC unit cell with an atomic radius of 126 pm. Calculate the edge length of the unit cell in pm.

ANSWER: ______________________________

112. How many atoms are contained within one face-centered cubic (FCC) unit cell?

ANSWER: ______________________________

113. Rank the following substances from highest to lowest vapor pressure at room temperature: H_2O, He, CO_2, N_2. For credit you must justify your answer.

ANSWER: ______________________________

114. Aluminum crystallizes in a FCC unit cell with an atomic radius of 143 pm. Calculate the edge length of the unit cell in pm.

ANSWER: ______________________________

115. Liquid O_2 at a constant 1.0 atm is heated from $-210\,°C$ to $-150\,°C$. Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: ______________________________

116. An unknown solid has the following properties: low melting point, soft, does not conduct electricity, volatile. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: ______________________________

117. Rank the following substances from lowest to highest vapor pressure at room temperature: N_2, CO_2, H_2O, He. For credit you must justify your answer.

ANSWER: ______________________________

118. A solution contains a crystal is added and rapid crystallization occurs at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: ______________________________

119. A solution of HCl (HCl) is 25.0% by mass. Calculate the molality of the solution. ($\mathscr{M}(HCl) = 36.46\,\text{g/mol}$)

ANSWER: ______________________________

120. Use the Born-Haber cycle to calculate ΔH_f° of NaF. $\Delta H_{\text{sub}}(\text{Na}) = 108\,\text{kJ/mol}$, $IE_1(\text{Na}) = 496\,\text{kJ/mol}$, $\frac{1}{2}D(\text{F}_2) = 79\,\text{kJ/mol}$, $EA(\text{F}) = -328\,\text{kJ/mol}$, $U(NaF) = -929\,\text{kJ/mol}$.

ANSWER: ______________________________

121. Classify a solid molecular sugar (sucrose) (sucrose ($C_{12}H_{22}O_{11}$)) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: ______________________________

122. Predict the shape of the meniscus formed by mercury in a glass tube. Justify using cohesive and adhesive forces.

ANSWER: ______________________________

123. A red blood cell (internal particle concentration $\approx 0.300\,\text{osmol/L}$) is placed in a $0.500\,\text{M}$ solution of $CaCl_2$ ($CaCl_2$). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: ______________________________

124. A system consists of pure water at its triple point. Determine its degrees of freedom using the Gibbs phase rule, then state what that result means. Justify your answer.

ANSWER: ______________________________

125. The lattice energy of LiCl (LiCl) is $+853\,\text{kJ/mol}$ (energy to separate ions) and the hydration energy is $-890\,\text{kJ/mol}$. Calculate ΔH_{sol}. Is dissolution endothermic or exothermic?

ANSWER: ______________________________

126. Calculate the boiling point of a solution prepared by dissolving $15.0\,\text{g}$ of urea (CH_4N_2O) in $250.\,\text{g}$ of water. $K_b(H_2O) = 0.512\,^\circ\text{C kg/mol}$.

ANSWER: ______________________________

127. For water (H_2O), $\Delta H_{\text{fus}} = 6.01$ kJ/mol and $\Delta H_{\text{vap}} = 40.7$ kJ/mol. Explain why boiling absorbs so much more energy than melting.

ANSWER: ______________________________

128. Determine the correct order of lowest to highest boiling point for the following substances: CS_2, CCl_4, $SiCl_4$, CO_2. Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: ______________________________

129. How many atoms are contained within one body-centered cubic (BCC) unit cell?

ANSWER: ______________________________

130. Rank the following ionic compounds from most negative (strongest) to least negative lattice energy: NaF, NaBr, NaCl.

ANSWER: ______________________________

131. Rank the following ionic compounds from least negative to most negative (strongest) lattice energy: LiF, KF, NaF.

ANSWER: ______________________________

132. On an intermolecular potential-energy curve, what does the depth of the potential well represent? Justify.

ANSWER: ______________________________

133. The following information is available for iron. Determine the density of iron and justify your answer.
atomic radius = 126 pm
unit-cell edge length = 286.65 pm
melting point = 1538 °C
molar mass = 55.845 $\mathrm{g\,mol^{-1}}$
packing efficiency = 68%
atoms per unit cell = 2
electron configuration = $[\mathrm{Ar}]3d^6 4s^2$

ANSWER: ______________________________

134. Methanol (CH_3OH) has a normal boiling point of 65°C and $\Delta H_{\mathrm{vap}} = 35.2$ kJ/mol. At what temperature does it boil inside a pressure cooker, where the external pressure is 2.14 atm? ($R = 8.314$ J $\mathrm{mol^{-1} K^{-1}}$.)

ANSWER: ______________________________

135. Calculate the osmotic pressure of a 0.0500 M aqueous solution of sucrose ($C_{12}H_{22}O_{11}$) at 25°C.

ANSWER: ______________________________

136. List every type of intermolecular force present in pure $CHCl_3$. Justify your answer.

ANSWER: ______________________

137. A sample of CO2 at 50 °C (above its critical temperature of 31 °C) is compressed to very high pressure. Will it ever separate into distinct liquid and gas phases? Justify.

ANSWER: ______________________

138. On an intermolecular potential-energy curve $u(r)$, what is the physical significance of the separation r at which $u(r)$ reaches its minimum? Justify.

ANSWER: ______________________

139. A solution of O_2 (O_2) in water at 25°C is prepared at a partial pressure of 0.50 atm. If the partial pressure is increased to 2.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: ______________________

140. Silver crystallizes in a FCC unit cell with an atomic radius of 144 pm. Calculate the edge length of the unit cell in pm.

ANSWER: ______________________

141. Solid krypton (Kr) has its triple point at -157°C and 0.72 atm. It is held at a constant external pressure of 0.129 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: ______________________

142. Determine the correct order of lowest to highest boiling point for the following substances: CH_3SH, CH_4, CH_3OH, CH_3Cl. Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: ______________________

143. Classify magnesium chloride ($MgCl_2$) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________

144. In an ionic crystal a cation (Na^+, radius 102 pm) sits among anions (Cl^-, radius 181 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: ______________________

145. A solution contains a crystal is added and rapid crystallization occurs at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: ______________________________

146. A 0.500 M solution of HCl (HCl) has a density of 1.02 g/mL. Calculate the molality of the solution. ($\mathscr{M}$ = 36.46 g/mol)

ANSWER: ______________________________

147. The enthalpy of solution of NaOH (NaOH) is $\Delta H_{sol} = -44.5$ kJ/mol. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when NaOH dissolves in water?

ANSWER: ______________________________

148. What is the coordination number of each atom in a hexagonal close-packed (HCP) structure?

ANSWER: ______________________________

149. A solution of ethanol and hexane is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

150. Which liquid is more viscous: ethylene glycol ($C_2H_6O_2$) or water (H_2O)? Justify your answer.

ANSWER: ______________________________

151. Predict how the viscosity of maple syrup changes as it is warmed. Justify your answer.

ANSWER: ______________________________

152. Calculate the mole fractions of acetone (C_3H_6O) and water in a solution prepared by mixing 6.40 g of acetone with 100. g of water.

ANSWER: ______________________________

153. The normal boiling point of methanol (CH_3OH) is 64.60°C ($P_{vap} = 1.000$ atm) and its molar enthalpy of vaporization is 35.2 kJ/mol. Calculate the vapor pressure of methanol at 42.60°C.

ANSWER: ______________________

154. Calculate the osmotic pressure of 0.90% saline at 25°C (modeled as a 0.154 M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: ______________________

155. Calculate the freezing point of a solution prepared by dissolving 20.0 g of glucose ($C_6H_{12}O_6$) in 100. g of water. $K_f(H_2O) = 1.86$°C kg/mol.

ANSWER: ______________________

156. For iodine (I_2): normal melting point 114°C, normal boiling point 184°C, triple point (114°C, 0.117 atm), critical point (546°C, 116 atm). What phase is present at 81°C and 1 atm? Justify your answer.

ANSWER: ______________________

157. Rank the following liquids from highest to lowest surface tension at 20°C: C_6H_6, C_6H_{14}, $C_3H_8O_3$, H_2O. Offer brief explanation for your ranking.

ANSWER: ______________________

158. Calculate the osmotic pressure of seawater at 25°C (modeled as a 0.600 M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: ______________________

159. A solution of glucose ($C_6H_{12}O_6$) is 10.0% by mass. Calculate the molality of the solution. ($\mathscr{M}(glucose) = 180.16$ g/mol)

ANSWER: ______________________

160. At 25°C and 1.00 atm, the solubility of N_2 (N_2) in water is 6.1×10^{-4} mol/L. What partial pressure of N_2 is required to achieve a solubility of 1.8×10^{-3} mol/L? Use Henry's law: $P = C/K_{\mathrm{H}}$.

ANSWER: ______________________________

161. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: ______________________________

162. Rank the following liquids from highest to lowest surface tension at 20°C: Hg, C_6H_{14}, CH_4, H_2O. Offer brief explanation for your ranking.

ANSWER: ______________________________

163. At 25°C and 1.00 atm, the solubility of N_2 (N_2) in water is 6.1×10^{-4} mol/L. What partial pressure of N_2 is required to achieve a solubility of 3.0×10^{-3} mol/L? Use Henry's law: $P = C/K_{\mathrm{H}}$.

ANSWER: ______________________________

164. The vapor pressure of diethyl ether ($C_4H_{10}O$) is 0.2869 atm at 0.00°C and 1.000 atm at 34.50°C. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________________

165. Calculate the molar solubility of CH_4 (CH_4) in water at 25°C when the partial pressure of CH_4 above the solution is 0.50 atm. Use $K_{\mathrm{H}} = 1.4 \times 10^{-3}$ mol/(L·atm) (Henry's law: $C = K_{\mathrm{H}}P$).

ANSWER: ______________________________

166. The molar enthalpy of vaporization of ethanol (C_2H_5OH) is 38.6 kJ/mol and $M = 46.068$ g/mol. Find the mass enthalpy of vaporization of ethanol.

ANSWER: ______________________________

167. What is the packing efficiency (percent of space occupied by atoms) of a body-centered cubic (BCC) unit cell?

ANSWER: ______________________________

168. A solution is prepared by dissolving 0.500 g of insulin in enough water to make 200. mL of solution. The osmotic pressure at 25°C is 0.0105 atm. Calculate the molar mass of insulin. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathscr{M} = (m/V)/M$.

ANSWER: ______________________________

169. Po crystallizes in a simple cubic structure with unit cell edge length 335.0 pm. Calculate the density of Po.

ANSWER: ______________________________

170. List every type of intermolecular force present in pure NH_3. Justify your answer.

ANSWER: ______________________________

171. When a narrow copper capillary tube is dipped into mercury, does the mercury rise or become depressed in the tube? Justify using cohesive and adhesive forces.

ANSWER: ______________________________

172. The vapor pressure of chloroform ($CHCl_3$) is 0.086 atm at 0.00°C and 1.000 atm at 61.00°C. Calculate ΔH_{vap} in kJ/mol. $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$.

ANSWER: ______________________________

173. An unknown solid has the following properties: low melting point, soft, does not conduct electricity, volatile. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: ______________________________

174. Equal masses (1.00 g each) of $CaCl_2$, Na_2SO_4, CH_3OH are each dissolved in 1.00 kg of water. Which solution has the greatest boiling point elevation?

ANSWER: ______________________________

175. A crystalline sample gives a diffraction maximum when illuminated by monochromatic X-rays using the conditions below. A crystal with plane spacing 427 pm produces the first-order diffraction maximum at a glancing angle of $\theta = 12.1°$. Determine the wavelength of the incident X-rays. Calculate the requested quantity and justify your answer.

ANSWER: ______________________________

176. A 24.4-g sample of ice at $-38°C$ absorbs 6.50 kJ of heat. Determine the final state and temperature. ($c_{ice} = 2.09$ J $g^{-1}°C^{-1}$, $\Delta H_{fus} = 334$ J/g.)

ANSWER: ______________________________

177. How many hydrogen bonds can one molecule of HF simultaneously form (donors + acceptors combined)?

ANSWER: ______________________________

178. How many grams of H_2O can freeze when 7.39-kJ of heat is released during freezing at 0°C? Assume $\Delta H = 334\,J/g$.

ANSWER: ______________________________

179. Why does the temperature stay constant along the melting plateau of a heating curve even though heat is continuously added? Justify.

ANSWER: ______________________________

180. Identify the dominant intermolecular force acting between acetone (CH_3COCH_3) and water (H_2O) when they are mixed. Justify your answer.

ANSWER: ______________________________

181. A solution shows a freezing point depression of 0.186 °C when 200. g of water is used as the solvent. How many grams of urea (CH_4N_2O, $\mathscr{M} = 60.06$ g/mol) are dissolved? ($K_f(H_2O) = 1.86$°C kg/mol)

ANSWER: ______________________

182. Describe the layer stacking sequence of hexagonal closest packing (HCP) and explain what makes it distinct. Justify.

ANSWER: ______________________

183. Two solutions are prepared: solution A is a 1.00 m solution of NaCl (NaCl) in water, and solution B is a 1.00 m solution of glucose ($C_6H_{12}O_6$) in water. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: ______________________

184. A red blood cell (internal particle concentration ≈ 0.300 osmol/L) is placed in a 0.150 M solution of Na_2SO_4 (Na_2SO_4). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: ______________________

185. A 0.500 M solution of glucose ($C_6H_{12}O_6$) has a density of 1.02 g/mL. Calculate the molality of the solution. ($\mathscr{M} = 180.16$ g/mol)

ANSWER: ______________________

186. The vapor pressure of chloroform ($CHCl_3$) is 0.2613 atm at 20.00°C and 1.000 atm at 61.00°C. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________

187. Which liquid has the higher surface tension: mercury (Hg) or water (H_2O)? Justify your answer.

ANSWER: ______________________

188. A solution contains more than the equilibrium maximum at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: ______________________

189. Rank the following ionic compounds from most negative (strongest) to least negative lattice energy: NaF, NaBr, NaCl.

ANSWER: ______________________________

190. Water (H_2O) has a normal boiling point of 100°C and $\Delta H_{\text{vap}} = 40.7$ kJ/mol. At what temperature does it boil at high altitude, where the external pressure is 0.7 atm? ($R = 8.314$ J mol^{-1}K^{-1}.)

ANSWER: ______________________________

191. Graphite and graphene are both composed entirely of carbon. Determine which would be expected to function better as a dry lubricant. State your answer and justify it.

ANSWER: ______________________________

192. In the cesium chloride (CsCl) structure (for example CsCl), there is a simple cubic array of anions with the cation in the cubic (body-center) hole. State the coordination numbers (cation:anion). Justify.

ANSWER: ______________________________

193. When sodium hydroxide (NaOH) dissolves in water the solution warms up. A student says this warming alone proves the dissolution is spontaneous. Is that reasoning sound? Explain.

ANSWER: ______________________________

194. Rank the following substances from highest to lowest vapor pressure at room temperature: He, H_2O, CO_2, N_2. For credit you must justify your answer.

ANSWER: ______________________________

195. Calculate the heat required to warm 11.5-g of ice from −5°C to 0°C. $c_{\text{ice}} = 2.09\,\text{J g}^{-1}{}^{\circ}\text{C}^{-1}$.

ANSWER: ______________________________

196. A solution contains a crystal is added and nothing happens at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: ______________________________

197. Two solutions are prepared: solution A is a 1.00 m solution of NaCl (NaCl) in water, and solution B is a 1.00 m solution of glucose ($C_6H_{12}O_6$) in water. Which solution has the greater freezing point depression? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: ____________________

198. Explain why a detergent still cleans effectively in hard water while ordinary soap forms a scum. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: ____________________

199. Calculate the molar solubility of O_2 (O_2) in water at 25°C when the partial pressure of O_2 above the solution is 2.0 atm. Use $K_H = 1.3 \times 10^{-3}$ mol/(L·atm) (Henry's law: $C = K_H P$).

ANSWER: ____________________

200. The molar enthalpy of vaporization of water (H_2O) is 40.7 kJ/mol and $M = 18.015$ g/mol. Find the mass enthalpy of vaporization of water.

ANSWER: ____________________

201. Rank the following noble gases from highest to lowest boiling point: He, Xe, Ne, Ar. All noble gases interact only via London dispersion forces.

ANSWER: ____________________

202. Rank the following noble gases from lowest to highest boiling point: Kr, Xe, He, Ne. All noble gases interact only via London dispersion forces.

ANSWER: ____________________

203. The vapor pressure of benzene (C_6H_6) is 0.2421 atm at 40.00°C and 1.000 atm at 80.10°C. Calculate the molar enthalpy of vaporization.

ANSWER: ____________________

204. List every type of intermolecular force present in pure CH_3F. Justify your answer.

ANSWER: ______________________________

205. Nickel crystallizes in a FCC unit cell with an atomic radius of 124 pm. Calculate the edge length of the unit cell in pm.

ANSWER: ______________________________

206. For ammonia (NH_3): normal melting point $-78°C$, normal boiling point $-33°C$, triple point ($-78°C$, 0.06 atm), critical point (132°C, 111 atm). What phase is present at $-115°C$ and 0.0086 atm? Justify your answer.

ANSWER: ______________________________

207. Rank the following equimolal aqueous solutions from smallest to largest freezing point depression: KCl, $AlCl_3$, $C_{12}H_{22}O_{11}$. All solutions have the same molality.

ANSWER: ______________________________

208. For nitrogen (N_2): normal melting point $-210°C$, normal boiling point $-196°C$, triple point ($-210°C$, 0.124 atm), critical point ($-147°C$, 34 atm). What phase is present at $-201°C$ and 1 atm? Justify your answer.

ANSWER: ______________________________

209. Rank the following noble gases from highest to lowest boiling point: Kr, Xe, He, Ne. All noble gases interact only via London dispersion forces.

ANSWER: ______________________________

210. Calculate the heat required to completely vaporize 60.7-g of ethanol (C_2H_5OH) at its normal boiling point of 78°C. $\Delta H_{vap} = 38.6\,kJ/mol$, $M = 46.068\,g/mol$.

ANSWER: ______________________________

211. Classify glucose ($C_6H_{12}O_6$) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

212. Predict the shape of the meniscus formed by kerosene in a glass tube. Justify using cohesive and adhesive forces.

ANSWER: ______________________________

213. Calculate the freezing point of a solution prepared by dissolving 10.0 g of glucose ($C_6H_{12}O_6$) in 100. g of water. $K_f(H_2O) = 1.86°C$ kg/mol.

ANSWER: ______________________________

214. When 3.60 g of an unknown nonelectrolyte is dissolved in 200. g of water (H_2O), the freezing point decreases by 0.558 °C. $K_f(water) = 1.86°C$ kg/mol. Calculate the molar mass of the unknown solute.

ANSWER: ______________________________

215. A solution of methanol and water is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

216. The vapor pressure of methanol (CH_3OH) is 0.130 atm at 20.00°C and 1.000 atm at 64.60°C. Calculate ΔH_{vap} in kJ/mol. $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$.

ANSWER: ______________________________

217. Two solutions are prepared: solution A is a 0.100 m solution of NaCl (NaCl) in water, and solution B is a 0.100 m solution of glucose ($C_6H_{12}O_6$) in water. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: ______________________________

218. An unknown metal crystallizes in a FCC structure with unit cell edge length 408.53 pm and density 10.51 g/cm^3. Calculate the molar mass of the metal and identify it.

ANSWER: ______________________________

219. Rank the following noble gases from highest to lowest boiling point: Ne, Ar, Xe, Kr. All noble gases interact only via London dispersion forces.

ANSWER: ______________________________

220. How does increasing pressure affect the solubility of CO_2 in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

221. A solution is prepared by dissolving 1.50 g of hemoglobin in enough water to make 50.0 mL of solution. The osmotic pressure at 25°C is 0.0114 atm. Calculate the molar mass of hemoglobin. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathscr{M} = (m/V)/M$.

ANSWER: ______________________________

222. Rank the following ionic compounds from most negative (strongest) to least negative lattice energy: NaBr, NaCl, NaF.

ANSWER: ______________________________

223. Ethanol (C_2H_5OH, density 0.790 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: ______________________________

224. Liquid O_2 at a constant 1.0 atm is heated from -210 °C to -150 °C. Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: ______________________________

225. Calculate the heat required to completely vaporize 36.4-g of ethanol (C_2H_5OH) at its normal boiling point of 78°C. $\Delta H_{vap} = 38.6$ kJ/mol, $M = 46.068$ g/mol.

ANSWER: ______________________________

226. On the *P–T* phase diagram of gallium (Ga), the solid–liquid boundary has a negative slope. Which is denser, the solid or the liquid? Justify.

ANSWER: ______________________________

227. How many atoms are contained within one face-centered cubic (FCC) unit cell?

ANSWER: ______________________________

228. Which substance has the lower vapour pressure at the same temperature: hydrogen sulfide (H_2S) or water (H_2O)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: ______________________________

229. You know a solution's molarity and need its molality. What additional property must you measure, and why is it required?

ANSWER: ______________________________

230. Identify the dominant intermolecular force acting between acetone (CH_3COCH_3) and water (H_2O) when they are mixed. Justify your answer.

ANSWER: ______________________________

231. Two isomers share the molecular formula C_5H_{12}: n-pentane (straight-chain (elongated)) and isopentane (branched). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: ______________________________

232. Rank the following liquids from highest to lowest surface tension at 20°C: C_6H_6, Hg, $(C_2H_5)_2O$, H_2O. Offer brief explanation for your ranking.

ANSWER: ______________________________

233. Rank the following equimolal aqueous solutions from smallest to largest boiling point elevation: CH_4N_2O, $MgBr_2$, Na_2SO_4. All solutions have the same molality.

ANSWER: ______________________________

234. The molar enthalpy of fusion of water (H_2O) is 6.01 kJ/mol and its molar enthalpy of vaporization is 40.7 kJ/mol. Calculate the molar enthalpy of sublimation.

ANSWER: ______________________

235. The mass enthalpy of vaporization of methanol (CH_3OH) is 1099 J/g and $M = 32.042$ g/mol. Find the molar enthalpy of vaporization of methanol.

ANSWER: ______________________

236. Equal masses (1.00 g each) of C_2H_5OH, $MgSO_4$, LiBr are each dissolved in 1.00 kg of water. Which solution has the greatest freezing point depression?

ANSWER: ______________________

237. Which substance evaporates more slowly under identical conditions: ethanol (C_2H_5OH) or octane (C_8H_{18})? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: ______________________

238. The molar enthalpy of vaporization of benzene (C_6H_6) is 30.8 kJ/mol and $M = 78.114$ g/mol. Find the mass enthalpy of vaporization of benzene.

ANSWER: ______________________

239. A close-packed metal has the layer stacking sequence ABAB, repeating. What close-packing structure is this, and what is its coordination number? Justify.

ANSWER: ______________________

240. How many grams of Fe can melt when 36.6-kJ of heat is absorbed during fusion at 1538°C? Assume $\Delta H = 247\,\mathrm{J/g}$.

ANSWER: ______________________

241. List every type of intermolecular force present in pure NH_3. Justify your answer.

ANSWER: ______________________________

242. The vapor pressure of ethanol (C_2H_5OH) is 0.2064 atm at 60.00°C and 1.000 atm at 78.40°C. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________________

243. When 3.03 g of an unknown nonelectrolyte is dissolved in 250. g of camphor (camphor), the freezing point decreases by 1.86 °C. $K_f(camphor) = 37.7°\text{C kg/mol}$. Calculate the molar mass of the unknown solute.

ANSWER: ______________________________

244. For pure carbon dioxide with solid and gas in equilibrium, determine how many intensive variables may be varied independently while the described condition is maintained. Justify your answer.

ANSWER: ______________________________

245. For benzene (C_6H_6), the solid is denser than the liquid. Does increasing the external pressure raise or lower its melting point? Justify.

ANSWER: ______________________________

246. When sodium hydroxide (NaOH) dissolves in water the solution warms up. A student says this warming alone proves the dissolution is spontaneous. Is that reasoning sound? Explain.

ANSWER: ______________________________

247. A solution shows a freezing point depression of 1.02 °C when 200. g of water is used as the solvent. How many grams of anthracene ($C_{14}H_{10}$, $\mathscr{M} = 178.23\,\text{g/mol}$) are dissolved? ($K_f(H_2O) = 1.86°\text{C kg/mol}$)

ANSWER: ______________________________

248. The molar enthalpy of sublimation of iodine (I_2) is 57.09 kJ/mol and its molar enthalpy of fusion is 15.52 kJ/mol. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________________

249. The lattice energy of NaF (NaF) is +929 kJ/mol (energy to separate ions) and the hydration energy is −904 kJ/mol. Calculate ΔH_{sol}. Is dissolution endothermic or exothermic?

ANSWER: ______________________________

250. When 23.6 g of an unknown nonelectrolyte is dissolved in 250. g of water (H_2O), the freezing point decreases by 0.512 °C. $K_f(water) = 1.86°\text{C kg/mol}$. Calculate the molar mass of the unknown solute.

ANSWER: ______________________________

251. When heated, sodium chloride melts sharply at one temperature, whereas window glass gradually softens over a range. Which is crystalline and which is amorphous? Justify.

ANSWER: ______________________________

252. A mixture contains particles with diameters of 0.5 nm. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

253. A solution of H_2 (H_2) in water at 25°C is prepared at a partial pressure of 1.0 atm. If the partial pressure is increased to 5.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: ______________________________

254. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.150 M CH_4N_2O, 0.150 M Na_2SO_4, 0.150 M NaCl. Justify each classification.

ANSWER: ______________________________

255. A plant root cell (internal particle concentration ≈ 0.300 osmol/L) is placed in a 0.0500 M solution of Na_2SO_4 (Na_2SO_4). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: ____________________

256. Predict what happens to solubility when KNO_3 is dissolved in hot water and then cooled. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: ____________________

257. How much heat is absorbed when 30.7-g of liquid water vaporizes at 100°C? $\Delta H_{vap} = 2260$ J/g.

ANSWER: ____________________

258. A solution shows a boiling point elevation of 0.512 °C when 200. g of water is used as the solvent. How many grams of naphthalene ($C_{10}H_8$, $\mathscr{M} = 128.17$ g/mol) are dissolved? ($K_b(H_2O) = 0.512$°C kg/mol)

ANSWER: ____________________

259. Liquid O_2 at a constant 1.0 atm is heated from −210 °C to −150 °C. Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: ____________________

260. A monochromatic X-ray beam strikes a crystalline solid. Successive crystal planes produce a strong reflected beam under the conditions given below. A crystal with plane spacing 196 pm produces the first-order diffraction maximum at a glancing angle of $\theta = 23.1°$. Determine the wavelength of the incident X-rays. Determine the missing quantity and justify your answer.

ANSWER: ____________________

261. Rank these carboxylic acids from most to least soluble in water: CH_3COOH, C_3H_7COOH, HCOOH, C_2H_5COOH. Explain the trend.

ANSWER: ____________________

262. The molar enthalpy of sublimation of water (H_2O) is 46.71 kJ/mol and its molar enthalpy of fusion is 6.01 kJ/mol. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________________

263. A mixture contains visible particles that settle on standing. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

264. A solution is prepared by dissolving 0.500 g of lysozyme in enough water to make 50.0 mL of solution. The osmotic pressure at 25°C is 0.0171 atm. Calculate the molar mass of lysozyme. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathscr{M} = (m/V)/M$.

ANSWER: ______________________________

265. A solution shows a freezing point depression of 0.186 °C when 100. g of water is used as the solvent. How many grams of urea (CH_4N_2O, $\mathscr{M} = 60.06$ g/mol) are dissolved? ($K_f(H_2O) = 1.86$°C kg/mol)

ANSWER: ______________________________

266. A system consists of pure benzene with solid and liquid in equilibrium. Determine its degrees of freedom using the Gibbs phase rule, then state what that result means. Justify your answer.

ANSWER: ______________________________

267. Acetic acid (CH_3COOH, density 1.05 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: ______________________________

268. Diamond and graphite are both pure carbon. Determine which material would be expected to conduct electricity more effectively under ordinary conditions. State your answer and justify it.

ANSWER: ______________________________

269. Will NaCl in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

270. How does increasing temperature affect the solubility of CO_2 in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

271. Identify the dominant type of intermolecular force present in pure CH_3OH. You must justify your answer.

ANSWER: ______________________________

272. For bismuth (Bi), the solid floats on its own liquid. What is the sign of the slope of its solid–liquid boundary on a P–T phase diagram? Justify.

ANSWER: ______________________________

273. Rank the following substances from highest to lowest vapor pressure at room temperature: He, H_2O, CO_2, N_2. For credit you must justify your answer.

ANSWER: ______________________________

274. Dissolving potassium nitrate (KNO_3) is endothermic ($\Delta H_{\mathrm{sol}} = +35.4\,\mathrm{kJ/mol}$), yet it happens spontaneously. What makes this possible? Explain.

ANSWER: ______________________________

275. A 0.500 M solution of HCl (HCl) has a density of 1.02 g/mL. Calculate the molality of the solution. ($\mathscr{M} = 36.46\,\mathrm{g/mol}$)

ANSWER: ______________________________

276. For nitrogen (N_2): normal melting point -210°C, normal boiling point -196°C, triple point (-210°C, 0.124 atm), critical point (-147°C, 34 atm). What phase is present at -178°C and 0.0248 atm? Justify your answer.

ANSWER: ______________________________

277. Describe the layer stacking sequence of cubic closest packing (CCP) and explain what makes it distinct. Justify.

ANSWER: ______________________________

278. Explain how a single crystal illuminated by monochromatic X-rays can produce many distinct diffraction peaks. Justify.

ANSWER: ______________________________

279. A solution of H_2 (H_2) in water at 25°C is prepared at a partial pressure of 2.0 atm. If the partial pressure is increased to 4.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: ______________________________

280. Explain why soap molecules only begin assembling into micelles above the critical micelle concentration. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: ______________________________

281. Rank these primary amines from most to least soluble in water: $C_3H_7NH_2$, $C_2H_5NH_2$, $C_8H_{17}NH_2$, $C_6H_{13}NH_2$. Explain the trend.

ANSWER: ______________________________

282. Rank the following noble gases from highest to lowest boiling point: Kr, He, Ne, Xe. All noble gases interact only via London dispersion forces.

ANSWER: ______________________________

283. Supercritical CO2 is used to decaffeinate coffee. Suggest why it is preferred over a liquid organic solvent, based on its physical properties.

ANSWER: ______________________________

284. When 11.1 g of an unknown nonelectrolyte is dissolved in 250. g of benzene (C_6H_6), the freezing point decreases by 1.86 °C. $K_f(benzene) = 5.12$°C kg/mol. Calculate the molar mass of the unknown solute.

ANSWER: ______________________________

285. The vapor pressure of diethyl ether ($C_4H_{10}O$) is 0.2869 atm at 0.00°C and 1.000 atm at 34.50°C. Calculate the molar enthalpy of vaporization.

ANSWER: ______________________________

286. A mixture contains visible particles that settle on standing. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

287. Predict what happens to solubility when a carbonated drink is warmed. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: ______________________________

288. Calculate the heat absorbed when 49.3-g of ethanol (C_2H_5OH) melts completely at its melting point of -114°C. $\Delta H_{\text{fus}} = 4.6\,\text{kJ/mol}$.

ANSWER: ______________________________

289. Graphite and graphene both consist entirely of carbon. Determine which material would be expected to conduct electricity more effectively within its carbon layer(s). State your answer and justify it.

ANSWER: ______________________________

290. When $7.09\,\text{g}$ of an unknown nonelectrolyte is dissolved in $250.\,\text{g}$ of benzene (C_6H_6), the freezing point decreases by $1.86\,°\text{C}$. $K_f(benzene) = 5.12°\text{C kg/mol}$. Calculate the molar mass of the unknown solute.

ANSWER: ______________________________

291. For pure iodine with liquid and vapor in equilibrium, determine how many intensive variables may be varied independently while the described condition is maintained. Justify your answer.

ANSWER: ______________________________

292. A beam of light is passed through saltwater. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

293. You know a solution's molarity and need its molality. What additional property must you measure, and why is it required?

ANSWER: ______________________________

294. For pure benzene with solid and liquid in equilibrium, determine how many intensive variables may be varied independently while the described condition is maintained. Justify your answer.

ANSWER: ______________________________

295. Calculate the total heat required to convert 25.4-g of ice at $-15°C$ to steam at $126°C$. Use: $c_{ice} = 2.09$, $c_{water} = 4.184$, $c_{steam} = 2.01\,J\,g^{-1}°C^{-1}$; $\Delta H_{fus} = 334\,J/g$, $\Delta H_{vap} = 2260\,J/g$.

ANSWER: ______________________________

296. For one mole of methanol (CH_3OH), which requires the most energy: melting, boiling, or subliming? Justify.

ANSWER: ______________________________

297. Will NaCl in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

298. You know a solution's molarity and need its molality. What additional property must you measure, and why is it required?

ANSWER: ______________________________

299. Chloroform ($CHCl_3$, density 1.49 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: ______________________________

300. How does increasing temperature affect the solubility of NaCl in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

301. Classify acetic acid (CH_3COOH) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

302. A solution shows a boiling point elevation of $1.02\,°C$ when 500. g of water is used as the solvent. How many grams of anthracene ($C_{14}H_{10}$, $\mathscr{M} = 178.23\,g/mol$) are dissolved? ($K_b(H_2O) = 0.512°C\,kg/mol$)

ANSWER: ______________________________

303. A solution is prepared by dissolving 20.0 g of ethylene glycol ($C_2H_6O_2$, $M = 62.07$ g/mol) in 100. g of water. Calculate the mole fraction of ethylene glycol.

ANSWER: ______________________________

304. Supercritical CO2 is used to decaffeinate coffee. Suggest why it is preferred over a liquid organic solvent, based on its physical properties.

ANSWER: ______________________________

305. Will AgCl in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

306. A solution of KCl (KCl) is 5.00% by mass. Calculate the molality of the solution. ($\mathscr{M}(KCl) = 74.55$ g/mol)

ANSWER: ______________________________

307. A mixture appears optically clear and does not scatter light. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

308. Predict how the viscosity of honey changes as it is warmed. Justify your answer.

ANSWER: ______________________________

309. A crystalline sample gives a diffraction maximum when illuminated by monochromatic X-rays using the conditions below. A crystal with plane spacing 488 pm produces the second-order diffraction maximum at a glancing angle of $\theta = 28.0°$. Determine the wavelength of the incident X-rays. Calculate the requested quantity and justify your answer.

ANSWER: ______________________________

310. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.0500 M $CaCl_2$, 0.100 M $MgCl_2$, 0.250 M CH_4N_2O. Justify each classification.

ANSWER: ______________________________

311. Rank the following liquids from highest to lowest surface tension at 20°C: $HOCH_2CH_2OH$, C_6H_{14}, H_2O, CH_3COCH_3. Offer brief explanation for your ranking.

ANSWER: ______________________________

312. A bacterium (internal particle concentration $\approx 0.300\,\text{osmol/L}$) is placed in a $0.0500\,\text{M}$ solution of Na_2SO_4 (Na_2SO_4). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: ______________________________

313. Two isomers share the molecular formula C_5H_{12}: n-pentane (straight-chain (elongated)) and isopentane (branched). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: ______________________________

314. Rank the following liquids from highest to lowest viscosity at 20°C: H_2O, corn syrup, olive oil, $C_2H_5OC_2H_5$.

ANSWER: ______________________________

315. Which substance has the lower vapour pressure at the same temperature: diethyl ether ($C_2H_5OC_2H_5$) or pentane (C_5H_{12})? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: ______________________________

316. A solution of ethanol and hexane is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

317. Calculate the boiling point of a solution prepared by dissolving 20.0 g of urea (CH_4N_2O) in 100. g of water. $K_b(H_2O) = 0.512°\text{C kg/mol}$.

ANSWER: ______________________________

318. Rank the following liquids from highest to lowest surface tension at 20°C: C_6H_{14}, $HOCH_2CH_2OH$, CH_3COCH_3, H_2O. Offer brief explanation for your ranking.

ANSWER: ______________________________

319. Two isomers share the molecular formula C_8H_{18}: n-octane (straight-chain (elongated)) and 2,2,4-trimethylpentane (highly branched (compact)). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: ______________________________

320. A heating curve has three sloped segments (solid, liquid, and gas regions) with different slopes. Why do the slopes differ? Justify.

ANSWER: ______________________________

321. A 52.8-g sample of ice at 0°C absorbs 13.9 kJ of heat. Determine the final state and temperature. ($\Delta H_{\text{fus}} = 334$ J/g.)

ANSWER: ______________________________

322. Rank the following substances from lowest to highest vapor pressure at room temperature: He, H_2O, CO_2, N_2. For credit you must justify your answer.

ANSWER: ______________________________

323. Without using memorized boiling-point trends, determine which substance has the higher boiling point: PH_3 or NH_3. State your answer and justify it.

ANSWER: ______________________________

324. Arrange the following substances from lowest to highest boiling point: CO_2, CCl_4, $SiCl_4$, CS_2. Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: ______________________________

325. When ammonium nitrate (NH_4NO_3) dissolves in water the solution turns cold, yet it dissolves on its own. A student concludes the process must be non-spontaneous. Is the student correct? Explain.

ANSWER: ______________________________

326. The normal boiling point of ethanol (C_2H_5OH) is 78.40°C ($P_{\text{vap}} = 1.000$ atm) and its molar enthalpy of vaporization is 38.6 kJ/mol. Calculate the vapor pressure of ethanol at 63.40°C.

ANSWER: ______________________________

327. Calculate the osmotic pressure of a 0.100 M aqueous solution of urea (CH_4N_2O) at 25°C.

ANSWER: ______________________________

328. Explain why X-rays, rather than visible light, are used to determine the spacing of atoms in crystals. Justify.

ANSWER: ______________________________

329. A 1.00 M solution of NaCl (NaCl) has a density of 1.05 g/mL. Calculate the molality of the solution. ($\mathscr{M} = 58.44$ g/mol)

ANSWER: ______________________________

330. Classify each as crystalline or amorphous: sodium chloride and window glass. Justify using their internal order.

ANSWER: ______________________________

331. Rank the following liquids from highest to lowest viscosity at 20°C: castor oil, CH_3COCH_3, H_2O, $HOCH_2CH_2OH$.

ANSWER: ______________________________

332. The molar enthalpy of vaporization of acetone (CH_3COCH_3) is 31.3 kJ/mol and $M = 58.08$ g/mol. Find the mass enthalpy of vaporization of acetone.

ANSWER: ______________________________

333. Explain why a detergent still cleans effectively in hard water while ordinary soap forms a scum. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: ______________________________

334. How many grams of glucose ($C_6H_{12}O_6$) must be dissolved in 200. g of water to raise the boiling point by 0.256 °C? $K_b(H_2O) = 0.512$°C kg/mol.

ANSWER: ______________________________

335. An unknown solid has a melting point of 1085°C, is a conductor as a solid, and is a conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: ______________________________

336. List every type of intermolecular force present in pure CH_3F. Justify your answer.

ANSWER: ______________________________

337. Explain how soap lifts greasy dirt into water so it can be rinsed away. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: ______________________________

338. An unknown solid has the following properties: very high melting point, extremely hard, does not conduct electricity. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: ______________________________

339. How many grams of urea (CH_4N_2O) must be dissolved in 200. g of water to raise the boiling point by 0.256 °C? $K_b(H_2O) = 0.512$°C kg/mol.

ANSWER: ______________________________

340. ZnS crystallizes in a zinc blende structure with unit cell edge length 541.0 pm. Calculate the density of ZnS.

ANSWER: ______________________________

341. Solid krypton (Kr) has its triple point at −157°C and 0.72 atm. It is held at a constant external pressure of 0.3612 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: ______________________________

342. Calculate the vapor pressure of a solution prepared by dissolving 54.0 g of glucose ($C_6H_{12}O_6$) in 180. g of water at 25°C. The vapor pressure of pure water at 25°C is 23.8 mmHg.

ANSWER: ______________________________

343. Identify the dominant intermolecular force acting between hexane (C_6H_{14}) and carbon tetrachloride (CCl_4) when they are mixed. Justify your answer.

ANSWER: ______________________________

344. Predict what happens to solubility when a lake is heated by a power plant. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: ______________________________

345. An unknown solid has a melting point of 1085°C, is a conductor as a solid, and is a conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: ______________________________

346. A solution is prepared by dissolving 25.0 g of sucrose ($C_{12}H_{22}O_{11}$, $M = 342.3$ g/mol) in 250. g of water. Calculate the molality of the solution.

ANSWER: ______________________________

347. An ionic solid has a face-centered cubic array of cations with anions in all of the tetrahedral holes. Name this structure type (with a common example) and justify.

ANSWER: ______________________________

348. A solution shows a freezing point depression of 0.186 °C when 200. g of water is used as the solvent. How many grams of naphthalene ($C_{10}H_8$, $\mathscr{M} = 128.17$ g/mol) are dissolved? ($K_f(H_2O) = 1.86$°C kg/mol)

ANSWER: ______________________________

349. How many hydrogen-bond acceptors does one molecule of HCOOH have?

ANSWER: ______________________________

350. Will oil in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

351. Can radiation of wavelength 194 pm produce a first-order Bragg reflection from crystal planes spaced 392 pm apart? Justify.

ANSWER: ______________________________

352. A 0.100 m aqueous solution of NaCl (NaCl) shows a freezing point depression of 0.335 °C ($K_f = 1.86$°C kg/mol). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: ______________________________

353. For bismuth (Bi), the solid floats on its own liquid. What is the sign of the slope of its solid–liquid boundary on a P–T phase diagram? Justify.

ANSWER: ______________________________

354. Use the data below to determine the density of aluminum. Justify your answer.
melting point = 660.3 °C
unit-cell edge length = 404.95 pm
molar mass = 26.982 g mol^{-1}
electron configuration = [Ne]$3s^2 3p^1$
atoms per unit cell = 4
crystal structure = FCC

ANSWER: ______________________________

355. The vapor pressure of diethyl ether ($(C_2H_5)_2O$) is 0.173 atm at -10.00°C and 1.000 atm at 34.50°C. Calculate ΔH_{vap} in kJ/mol. $R =$ 8.314 J mol^{-1}K^{-1}.

ANSWER: ______________________________

356. When 1.77 g of an unknown nonelectrolyte is dissolved in 200. g of benzene (C_6H_6), the freezing point decreases by 0.372 °C. $K_f(\textit{benzene}) =$ 5.12°C kg/mol. Calculate the molar mass of the unknown solute.

ANSWER: ______________________________

357. Two solutions are prepared: solution A is a 1.00 m solution of NaCl (NaCl) in water, and solution B is a 1.00 m solution of glucose ($C_6H_{12}O_6$) in water. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: ______________________________

358. At 25°C and 1.00 atm, the solubility of N_2 (N_2) in water is 6.1×10^{-4} mol/L. What partial pressure of N_2 is required to achieve a solubility of 1.2×10^{-3} mol/L? Use Henry's law: $P = C/K_H$.

ANSWER: ______________________________

359. For methanol (CH_3OH), $\Delta H_{fus} = 3.16$ kJ/mol and $\Delta H_{vap} = 35.2$ kJ/mol. Explain why boiling absorbs so much more energy than melting.

ANSWER: ______________________________

360. Classify a solid of magnesium and oxygen (MgO) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: ______________________________

361. In the zinc blende (ZnS) structure (for example ZnS), there is a face-centered cubic array of anions with cations in half of the tetrahedral holes. State the coordination numbers (cation:anion). Justify.

ANSWER: ______________________________

362. A beam of light is passed through milk. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

363. A 0.250 m aqueous solution of urea (CH_4N_2O) shows a freezing point depression of 0.465 °C ($K_f = 1.86$°C kg/mol). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: ______________________________

364. At 25°C and 1.00 atm, the solubility of O_2 (O_2) in water is 1.3×10^{-3} mol/L. What partial pressure of O_2 is required to achieve a solubility of 6.5×10^{-3} mol/L? Use Henry's law: $P = C/K_H$.

ANSWER: ______________________________

365. Identify the dominant intermolecular force acting between ammonia (NH_3) and water (H_2O) when they are mixed. Justify your answer.

ANSWER: ______________________________

366. How many grams of sucrose ($C_{12}H_{22}O_{11}$) must be dissolved in 100. g of water to raise the boiling point by 0.256 °C? $K_b(H_2O) = 0.512°C\,kg/mol$.

ANSWER: ____________________

367. How many hydrogen-bond acceptors does one molecule of NH_3 have?

ANSWER: ____________________

368. Will NH_3 in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: ____________________

369. Water (H_2O) has a normal boiling point of 100°C and $\Delta H_{vap} = 40.7$ kJ/mol. At what temperature does it boil at high altitude, where the external pressure is 0.5 atm? ($R = 8.314$ J mol^{-1}K^{-1}.)

ANSWER: ____________________

370. A semipermeable membrane separates a 0.10 M sucrose solution from a 0.25 M sucrose solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ____________________

371. The enthalpy of solution of glucose ($C_6H_{12}O_6$) is $\Delta H_{sol} = +10.9\,kJ/mol$. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when glucose dissolves in water?

ANSWER: ____________________

372. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: ____________________

373. A 0.0500 m aqueous solution of $MgCl_2$ ($MgCl_2$) shows a boiling point elevation of 0.0691 °C ($K_b = 0.512$°C kg/mol). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: ____________________

374. Calculate the vapor pressure of a solution prepared by dissolving 54.0 g of urea (CH_4N_2O) in 180. g of water at 25°C. The vapor pressure of pure water at 25°C is 23.8 mmHg.

ANSWER: ____________________

375. An unknown solid has a melting point of −57°C, is a non-conductor as a solid, and is a non-conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: ____________________

376. The lattice energy of CsI (CsI) is +604 kJ/mol (energy to separate ions) and the hydration energy is −567 kJ/mol. Calculate ΔH_{sol}. Is dissolution endothermic or exothermic?

ANSWER: ____________________

377. A 0.100 m aqueous solution of KCl (KCl) shows a boiling point elevation of 0.0870 °C ($K_b = 0.512$°C kg/mol). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: ____________________

378. In an ionic crystal a cation (Cs^+, radius 167 pm) sits among anions (Cl^-, radius 181 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: ____________________

379. The molar enthalpy of sublimation of naphthalene ($C_{10}H_8$) is 62.36 kJ/mol and its molar enthalpy of vaporization is 43.3 kJ/mol. Calculate the molar enthalpy of fusion.

ANSWER: ____________________

380. Calculate the freezing point of a solution prepared by dissolving 10.0 g of NaCl (NaCl) in 500. g of water. $K_f(H_2O) = 1.86°C$ kg/mol. Assume complete dissociation ($i = 2$).

ANSWER: ______________________________

381. Identify the dominant type of intermolecular force present in pure C_6H_6. You must justify your answer.

ANSWER: ______________________________

382. Water (H_2O) has a normal boiling point of 100°C and $\Delta H_{vap} = 40.7$ kJ/mol. At what temperature does it boil at high altitude, where the external pressure is 0.8 atm? ($R = 8.314$ J mol^{-1}K^{-1}.)

ANSWER: ______________________________

383. On a heating curve of temperature versus heat added, one segment is a flat plateau at the boiling point. What is happening to the substance there, and which equation gives the heat? Justify.

ANSWER: ______________________________

384. A solution contains a crystal is added and nothing happens at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: ______________________________

385. How many hydrogen-bond donors does one molecule of CH_3COOH have?

ANSWER: ______________________________

386. On a heating curve, a rising (sloped) segment lies between the melting and boiling plateaus. What is happening, and which equation applies? Justify.

ANSWER: ______________________________

387. A solution contains a crystal is added and nothing happens at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: ______________________________

388. Ethanol (C_2H_5OH) has a normal boiling point of 78°C and $\Delta H_{vap} = 38.6$ kJ/mol. At what temperature does it boil at high altitude, where the external pressure is 0.8 atm? ($R = 8.314$ J mol^{-1}K^{-1}.)

ANSWER: ____________________

389. A mixture appears optically clear and does not scatter light. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ____________________

390. Explain why a detergent still cleans effectively in hard water while ordinary soap forms a scum. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: ____________________

391. Acetic acid (CH_3COOH, density 1.05 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: ____________________

392. A 78.9-g sample of ice at −30°C absorbs 21.7 kJ of heat. Determine the final state and temperature. ($c_{ice} = 2.09$ J g^{-1}°C^{-1}, $\Delta H_{fus} = 334$ J/g.)

ANSWER: ____________________

393. Which liquid has the higher surface tension: ethanol (C_2H_5OH) or hexane (C_6H_{14})? Justify your answer.

ANSWER: ____________________

394. Rank the following liquids from highest to lowest viscosity at 20°C: H_2O, C_2H_5OH, $C_3H_8O_3$, C_6H_{14}.

ANSWER: ____________________

395. The enthalpy of solution of KNO_3 (KNO_3) is $\Delta H_{sol} = +35.4$ kJ/mol. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when KNO_3 dissolves in water?

ANSWER: ____________________

396. Rank these carboxylic acids from most to least soluble in a nonpolar solvent such as hexane: $HCOOH$, $C_5H_{11}COOH$, CH_3COOH, C_3H_7COOH. Explain the trend.

ANSWER: ______________________________

397. A solution of N_2 (N_2) in water at 25°C is prepared at a partial pressure of 0.50 atm. If the partial pressure is increased to 1.5 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: ______________________________

398. How many grams of sucrose ($C_{12}H_{22}O_{11}$) must be dissolved in 1.00×10^3 g of water to raise the boiling point by 0.256 °C? $K_b(H_2O) = 0.512$°C kg/mol.

ANSWER: ______________________________

399. Classify paint as a type of colloid: identify its dispersed phase and dispersion medium, then name the colloid type.

ANSWER: ______________________________

400. Predict the shape of the meniscus formed by water in a glass tube. Justify using cohesive and adhesive forces.

ANSWER: ______________________________

401. Rank the following equimolal aqueous solutions from smallest to largest freezing point depression: $CaCl_2$, $NaCl$, $C_6H_{12}O_6$. All solutions have the same molality.

ANSWER: ______________________________

402. How many atoms are contained within one simple cubic (SC) unit cell?

ANSWER: ______________________________

403. A beam of light is passed through smoke. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

404. Classify a solid made of discrete nonpolar molecules (dry ice) (CO_2) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: ______________________________

405. An unknown metal crystallizes in a FCC structure with unit cell edge length 361.49 pm and density 8.94 g/cm^3. Calculate the molar mass of the metal and identify it.

ANSWER: ______________________________

406. Calculate the osmotic pressure of 0.90% saline at 25°C (modeled as a 0.154 M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: ______________________________

407. Classify marshmallow as a type of colloid: identify its dispersed phase and dispersion medium, then name the colloid type.

ANSWER: ______________________________

408. Rank the following liquids from highest to lowest viscosity at 20°C: $C_3H_8O_3$, C_2H_5OH, syrup, gasoline.

ANSWER: ______________________________

409. Rank the following equimolal aqueous solutions from smallest to largest boiling point elevation: KCl, $C_{12}H_{22}O_{11}$, $AlCl_3$. All solutions have the same molality.

ANSWER: ______________________________

410. Calculate the osmotic pressure of a 0.500 M aqueous solution of urea (CH_4N_2O) at 25°C.

ANSWER: ______________________________

411. At 25°C and 1.00 atm, the solubility of N_2 (N_2) in water is 6.1×10^{-4} mol/L. What partial pressure of N_2 is required to achieve a solubility of 2.4×10^{-3} mol/L? Use Henry's law: $P = C/K_H$.

ANSWER: ______________________________

412. A solution is prepared by dissolving 50.0 g of ethylene glycol ($C_2H_6O_2$, $M = 62.07$ g/mol) in 200. g of water. Calculate the molality of the solution.

ANSWER: ______________________________

413. The vapor pressure of acetone (CH_3COCH_3) is 0.096 atm at 0.00°C and 1.000 atm at 56.00°C. Calculate ΔH_{vap} in kJ/mol. $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$.

ANSWER: ______________________________

414. An unknown solid has the following properties: high melting point, brittle, conducts when molten, does not conduct as solid. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: ______________________________

415. An unknown metal crystallizes in a FCC structure with unit cell edge length 404.95 pm and density 2.7 g/cm^3. Calculate the molar mass of the metal and identify it.

ANSWER: ______________________________

416. A solution of heptane and octane is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

417. Identify the dominant intermolecular force acting between ammonia (NH_3) and water (H_2O) when they are mixed. Justify your answer.

ANSWER: ______________________________

418. For benzene (C_6H_6), $\Delta H_{\text{fus}} = 9.87$ kJ/mol and $\Delta H_{\text{vap}} = 30.8$ kJ/mol. Explain why boiling absorbs so much more energy than melting.

ANSWER: ______________________________

419. How many atoms are contained within one simple cubic (SC) unit cell?

ANSWER: ______________________________

420. How many grams of glucose ($C_6H_{12}O_6$) must be dissolved in 1.00×10^3 g of water to raise the boiling point by 0.512 °C? $K_b(H_2O) = 0.512°\text{C kg/mol}$.

ANSWER: ______________________________

421. Determine the correct order of lowest to highest boiling point for the following substances: CH_4, C_3H_8, C_4H_{10}, C_2H_6. Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: ______________________________

422. A solution is prepared by dissolving 25.0 g of glucose ($C_6H_{12}O_6$, $M = 180.16$ g/mol) in 500. g of water. Calculate the mole fraction of glucose.

ANSWER: ______________________________

423. For a sample of pure liquid ethanol, determine how many intensive variables may be varied independently while the described condition is maintained. Justify your answer.

ANSWER: ______________________________

424. Identify the dominant type of intermolecular force present in pure Br_2. You must justify your answer.

ANSWER: ______________________________

425. Calculate the vapor pressure of a solution prepared by dissolving 18.0 g of sucrose ($C_{12}H_{22}O_{11}$) in 200. g of water at 80°C. The vapor pressure of pure water at 80°C is 355.1 mmHg.

ANSWER: ______________________________

426. Use the data below to determine the density of silver. Justify your answer.
molar mass = $107.868\,\mathrm{g\,mol^{-1}}$
coordination number = 12
atomic radius = 144 pm
crystal structure = FCC
unit-cell edge length = 408.53 pm

ANSWER: ______________________________

427. A semipermeable membrane separates a 0.15 M glucose solution from a 0.15 M glucose solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

428. The ABCABC (repeating) close-packing sequence corresponds to which unit cell, and what fraction of space does it fill? Justify.

ANSWER: ______________________________

429. Calculate the mole fractions of ethanol (C_2H_5OH) and water in a solution prepared by mixing 18.4 g of ethanol with 100. g of water.

ANSWER: ______________________________

430. How many grams of CO_2 can sublime when 35.0-kJ of heat is absorbed during sublimation at -78.5°C? Assume $\Delta H = 571\,\mathrm{J/g}$.

ANSWER: ______________________________

431. Carbon tetrachloride (CCl_4, density 1.59 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: ______________________________

432. Equal masses (1.00 g each) of CH_4N_2O, KCl, $AlCl_3$ are each dissolved in 1.00 kg of water. Which solution has the greatest freezing point depression?

ANSWER: ______________________________

433. Calculate the freezing point of a solution prepared by dissolving 15.0 g of $MgSO_4$ ($MgSO_4$) in 100. g of water. $K_f(H_2O) = 1.86°C\,kg/mol$. Assume complete dissociation ($i = 2$).

ANSWER: ______________________

434. Identify the dominant type of intermolecular force present in pure Cl_2. You must justify your answer.

ANSWER: ______________________

435. The vapor pressure of diethyl ether ($(C_2H_5)_2O$) is 0.173 atm at $-10.00°C$ and 1.000 atm at 34.50°C. Calculate ΔH_{vap} in kJ/mol. $R = 8.314\,J\,mol^{-1}K^{-1}$.

ANSWER: ______________________

436. Calculate the mole fractions of acetone (C_3H_6O) and water in a solution prepared by mixing 12.8 g of acetone with 50.0 g of water.

ANSWER: ______________________

437. Use the Born-Haber cycle to calculate the lattice energy of LiF. $\Delta H^\circ_f(\text{LiF}) = -617\,\text{kJ/mol}$, $\Delta H_{sub}(\text{Li}) = 159\,\text{kJ/mol}$, $IE_1(\text{Li}) = 520\,\text{kJ/mol}$, $\frac{1}{2}D(F_2) = 79\,\text{kJ/mol}$, $EA(\text{F}) = -328\,\text{kJ/mol}$.

ANSWER: ______________________

438. An unknown solid has the following properties: variable melting point, malleable/ductile, conducts electricity as solid. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: ______________________

439. Use the Born-Haber cycle to calculate the lattice energy of LiF. $\Delta H^\circ_f(\text{LiF}) = -617\,\text{kJ/mol}$, $\Delta H_{sub}(\text{Li}) = 159\,\text{kJ/mol}$, $IE_1(\text{Li}) = 520\,\text{kJ/mol}$, $\frac{1}{2}D(F_2) = 79\,\text{kJ/mol}$, $EA(\text{F}) = -328\,\text{kJ/mol}$.

ANSWER: ______________________

440. Solid carbon dioxide (CO_2) has its triple point at $-56.6°C$ and 5.11 atm. It is held at a constant external pressure of 2.527 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: ______________________________

441. Calculate the freezing point of a solution prepared by dissolving 20.0 g of urea (CH_4N_2O) in 250. g of water. $K_f(H_2O) = 1.86°C\,kg/mol$.

ANSWER: ______________________________

442. Rank the following ionic compounds from least negative to most negative (strongest) lattice energy: LiCl, LiBr, LiI, LiF.

ANSWER: ______________________________

443. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: ______________________________

444. The mass enthalpy of vaporization of benzene (C_6H_6) is 394 J/g and $M = 78.114\,g/mol$. Find the molar enthalpy of vaporization of benzene.

ANSWER: ______________________________

445. A solution of methanol and water is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

446. In the fluorite (CaF2) structure (for example CaF_2), there is a face-centered cubic array of cations with anions in all of the tetrahedral holes. State the coordination numbers (cation:anion). Justify.

ANSWER: ______________________________

447. How does increasing pressure affect the solubility of NaCl in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

448. A semipermeable membrane separates pure water from a 0.1 M NaCl solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

449. Classify each as crystalline or amorphous: a quartz crystal and rubber. Justify using their internal order.

ANSWER: ______________________________

450. Calculate the mole fractions of ethanol (C_2H_5OH) and water in a solution prepared by mixing 12.8 g of ethanol with 90.0 g of water.

ANSWER: ______________________________

451. Solid water (H_2O) has its triple point at 0.01°C and 0.006 atm. It is held at a constant external pressure of 0.025 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: ______________________________

452. A 37.6-g sample of ice at 0°C absorbs 17.7 kJ of heat. Determine the final state and temperature. ($\Delta H_{\text{fus}} = 334$ J/g, $c_{\text{water}} = 4.18$ J g^{-1}°C^{-1}.)

ANSWER: ______________________________

453. On an intermolecular potential-energy curve $u(r)$, what is the physical significance of the separation r at which $u(r)$ reaches its minimum? Justify.

ANSWER: ______________________________

454. A semipermeable membrane separates a 0.05 M glucose solution from a 0.20 M glucose solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

455. Which concentration unit is a dimensionless ratio of particle counts, free of any volume or temperature dependence, and where is that property most useful?

ANSWER: ______________________________

456. Liquid O_2 at a constant 1.0 atm is heated from $-210\ °C$ to $-150\ °C$. Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: ______________________________

457. A 0.500 M solution of glucose ($C_6H_{12}O_6$) has a density of 1.02 g/mL. Calculate the molality of the solution. ($\mathscr{M} = 180.16$ g/mol)

ANSWER: ______________________________

458. One of H_2O and H_2S has a substantially higher boiling point than the other. Determine which one and justify your answer.

ANSWER: ______________________________

459. NaCl crystallizes in a rock salt structure with unit cell edge length 564.0 pm. Calculate the density of NaCl.

ANSWER: ______________________________

460. Predict what happens to solubility when KNO_3 is dissolved in hot water and then cooled. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: ______________________________

461. A phase diagram's liquid–gas curve ends at the critical point, but the solid–liquid curve never ends. Explain why there is no solid–liquid critical point.

ANSWER: ____________________

462. A beam of light is passed through gelatin. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ____________________

463. An unknown metal crystallizes in a FCC structure with unit cell edge length 404.95 pm and density 2.7 g/cm^3. Calculate the molar mass of the metal and identify it.

ANSWER: ____________________

464. Predict what happens to solubility when deep-sea fish are brought to the surface quickly. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: ____________________

465. Two isomers share the molecular formula C_4H_{10}: isobutane (branched (compact)) and n-butane (straight-chain). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: ____________________

466. Classify NaCl (NaCl) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ____________________

467. Calculate the heat required to warm 39.8-g of ice from -22°C to 0°C. $c_{\text{ice}} = 2.09\,\text{J g}^{-1}\,°\text{C}^{-1}$.

ANSWER: ____________________

468. Liquid O_2 at a constant 1.0 atm is heated from -210 °C to -150 °C. Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: ____________________

469. Two pure substances, NH_3 and PH_3, are compared. Predict which requires the higher temperature to boil. Justify your prediction. Unsupported trend statements receive no credit.

ANSWER: ______________________

470. A 0.100 m aqueous solution of $MgCl_2$ ($MgCl_2$) shows a freezing point depression of 0.558 °C ($K_f = 1.86$°C kg/mol). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: ______________________

471. The enthalpy of solution of ammonium nitrate (NH_4NO_3) is $\Delta H_{\text{sol}} = +25.7$ kJ/mol. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when ammonium nitrate dissolves in water?

ANSWER: ______________________

472. An unknown solid has a melting point of 2852°C, is a non-conductor as a solid, and is a conductor when molten. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: ______________________

473. A crystal produces an intense diffraction maximum for monochromatic X-rays under the conditions below. Monochromatic X-rays of wavelength 154 pm produce the first-order diffraction maximum at a glancing angle of $\theta = 10.3°$. Determine the spacing between adjacent crystal planes. Calculate the requested quantity and justify your answer.

ANSWER: ______________________

474. Rank these primary amines from most to least soluble in water: $C_3H_7NH_2$, $C_4H_9NH_2$, $C_8H_{17}NH_2$, $C_6H_{13}NH_2$. Explain the trend.

ANSWER: ______________________

475. Solid sulfur dioxide (SO_2) has its triple point at -75.5°C and 0.017 atm. It is held at a constant external pressure of 0.0051 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: ______________________

476. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: ______________________

477. In an ionic crystal a cation (Na^+, radius 102 pm) sits among anions (Cl^-, radius 181 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: ______________________________

478. Calculate the boiling point of a solution prepared by dissolving 10.0 g of $CaCl_2$ ($CaCl_2$) in 200. g of water. $K_b(H_2O) = 0.512°C\,kg/mol$. Assume complete dissociation ($i = 3$).

ANSWER: ______________________________

479. Use the Born-Haber cycle to calculate the lattice energy of KCl. $\Delta H_f^\circ(KCl) = -437\,kJ/mol$, $\Delta H_{sub}(K) = 89\,kJ/mol$, $IE_1(K) = 419\,kJ/mol$, $\frac{1}{2}D(Cl_2) = 121\,kJ/mol$, $EA(Cl) = -349\,kJ/mol$.

ANSWER: ______________________________

480. Will NaCl in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________

481. How much heat is absorbed when 21.1-g of liquid water vaporizes at 100°C? $\Delta H_{vap} = 2260\,J/g$.

ANSWER: ______________________________

482. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.100 M $CaCl_2$, 0.250 M CH_4N_2O, 0.150 M NaCl. Justify each classification.

ANSWER: ______________________________

483. In an ionic crystal a cation (Cs^+, radius 167 pm) sits among anions (Cl^-, radius 181 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: ______________________________

484. An unknown solid has a melting point of 114°C, is a non-conductor as a solid, and is a non-conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: ______________________________

485. The molar enthalpy of fusion of water (H_2O) is 6.01 kJ/mol and its molar enthalpy of vaporization is 40.7 kJ/mol. Calculate the molar enthalpy of sublimation.

ANSWER: ______________________________

486. The enthalpy of solution of ammonium nitrate (NH_4NO_3) is $\Delta H_{sol} = +25.7$ kJ/mol. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when ammonium nitrate dissolves in water?

ANSWER: ______________________________

487. The lattice energy of LiCl (LiCl) is $+853$ kJ/mol (energy to separate ions) and the hydration energy is -890 kJ/mol. Calculate ΔH_{sol}. Is dissolution endothermic or exothermic?

ANSWER: ______________________________

488. Calculate the molar solubility of O_2 (O_2) in water at 25°C when the partial pressure of O_2 above the solution is 4.0 atm. Use $K_H = 1.3 \times 10^{-3}$ mol/(L·atm) (Henry's law: $C = K_H P$).

ANSWER: ______________________________

489. Explain why a supercritical fluid can dissolve solutes like a liquid yet diffuse through a solid matrix like a gas.

ANSWER: ______________________________

490. Calculate the osmotic pressure of brackish water at 25°C (modeled as a 0.0500 M electrolyte solution with $i = 1$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: ______________________________

491. A colloid consists of a liquid dispersed in a liquid. Name this class of colloid and give an everyday example.

ANSWER: ______________________________

492. Dissolving ammonium nitrate (NH_4NO_3) is endothermic ($\Delta H_{sol} = +25.7\,kJ/mol$), yet it happens spontaneously. What makes this possible? Explain.

ANSWER: ______________________________

493. Rank the following liquids from highest to lowest viscosity at 20°C: CH_3OH, C_5H_{12}, C_3H_7OH, $HOCH_2CH_2OH$.

ANSWER: ______________________________

494. Rank these alcohols from most to least soluble in water: $C_8H_{17}OH$, $C_6H_{13}OH$, C_4H_9OH, $C_5H_{11}OH$. Explain the trend.

ANSWER: ______________________________

495. How many hydrogen-bond donors does one molecule of H_2O have?

ANSWER: ______________________________

496. Rank the following ionic compounds from least negative to most negative (strongest) lattice energy: LiF, KF, NaF.

ANSWER: ______________________________

497. Equal masses (1.00 g each) of $C_6H_{12}O_6$, Na_2CO_3, $Ca(NO_3)_2$ are each dissolved in 1.00 kg of water. Which solution has the greatest freezing point depression?

ANSWER: ______________________________

498. Calculate the boiling point of a solution prepared by dissolving 20.0 g of sucrose ($C_{12}H_{22}O_{11}$) in 200. g of water. $K_b(H_2O) = 0.512°C\,kg/mol$.

ANSWER: ______________________________

499. Which liquid is more viscous: glycerol ($C_3H_8O_3$) or ethanol (C_2H_5OH)? Justify your answer.

ANSWER: ______________________________

500. A semipermeable membrane separates pure water from a 0.50 M glucose solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: ______________________________